Protecting Wildlife at Scale: An Optimization Framework for Ranger–Technology Coordinated Patrols in Etosha National Park

Team Control Number: 2026000

Problem

Large reserves such as Etosha National Park (22,935 km², 295 staff) cannot be monitored everywhere at once. The IMMC asks how limited rangers and technology should be coordinated across space and time to provide measurable, strategically deployed protection — and how many personnel this requires.

Approach

We discretize the park into a 12×12 grid (~159 km² per cell) and define a quantifiable **Protection Index**: per-cell protection is a saturating (concave) function $E_i = 1 - e^{-u_i}$ of the effective surveillance intensity u_i built from six resource types (foot patrols, vehicle patrols, fixed sensors, drone units, satellite imagery, AI analytics), each with terrain-dependent effectiveness. A mixed-integer program maximizes the priority-weighted park-wide protection index **P** subject to budget, personnel (with a 72% duty-availability factor), equipment ceilings, and deployment-span constraints. A seasonal threat model extends the analysis over 12 months; a budgeted uncertainty set quantifies robustness; sensitivity and cross-park adaptation complete the study.

Key Results

- **Protection is definable and measurable.** We define "protected" as a cell-level probability of detection/interception exceeding 63% ($u_i \geq 1$), aggregated into a single park-wide index $P \in [0,1]$. Etosha's priority landscape is dominated by rhino-poaching pressure at the road-accessible savanna periphery.
- **Optimal baseline strategy:** $P = \mathbf{0.80}$ at a cost of **$675k/yr** using 156 field-effective staff. Coverage concentrates on high-priority cells and deliberately excludes the 4,800 km² salt pan — strategic deployment, not full coverage.
- **Personnel requirement:** protection rises steeply up to ≈150–210 field staff, then saturates because equipment capacity — not people — becomes the binding constraint. We recommend **≈155 operational rangers** plus re-tasking of the remaining of the 295 current staff toward

logistics, intelligence, and community liaison; the marginal value of an additional ranger falls from +0.007 P/person below 150 staff to ≈0 above 210.

- **Robustness:** if every cell's threat is 60% worse than nominal, protection retains 79% of baseline. Foot and vehicle patrols are the most critical assets (−0.088 and −0.066 P if lost); the model degrades gracefully under budget cuts (−70% budget ⇒ P = 0.63).
- **Portability:** the same structure transfers to Kaziranga (India) and Yellowstone (USA) by re-calibrating four difficulty parameters; required staffing multipliers are ×1.34 and ×1.54 respectively.

Table of Contents

1 Introduction and Problem Restatement 3
2 Assumptions and Justifications 4
3 Notation 4
4 Conservation Priorities and Defining "Protection" (Task 1) 5
5 Resource Allocation Model (Task 2) 7
6 Protection over Time and Staffing Needs (Task 3) 10
7 Sensitivity and Scenario Analysis (Task 4) 12
8 Model Strengths and Limitations (Task 5) 15
9 Adaptation to Other Protected Areas (Task 6) 16
10 Conclusions 18
Letter to the IMMC 19
Reference List 21
Appendix A: Computational Details 22
Report on Use of AI Tools 23

Team # 2026000
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Letter to the International Mission for Monitoring Conservation

Dear Members of the IMMC,

Etosha National Park is 22,935 square kilometers of savanna, salt pan, and thorn scrub, patrolled by 295 people. That is roughly one ranger for every 78 km² — the area of a small city — to protect one of Africa's last great black rhino populations. No organization can put a fence of wardens around such a landscape. The real question is therefore not "how do we watch everything?" but "where should each hour of patrol, each sensor, and each drone flight do the most good?" Our team built a mathematical tool that answers exactly that question, and we summarize what we learned.

What "protected" should mean

First, protection must be a number, not a hope. We measure the protection of any part of the park as the chance that a poacher or a fire starting there is detected and intercepted in time. A zone counts as protected when that chance exceeds about 63%; these scores are then weighted by ecological value and threat and averaged into a single Park Protection Index from 0 to 1. Etosha today sits near 0.55 under current practice; our plan raises it to about 0.80.

What to do with the money and the people

Our optimization recommends a layered defense: vehicle patrols along the road-permeable western and southern fringe where rhino poaching concentrates, foot patrols pushed into the high-value core, fixed camera/sensor nodes at the 86 waterholes where animals (and poachers) gather, drone units held as a rapid-reaction and night-capability reserve, with satellite imagery and an AI analytics platform stitching the picture together. Notably, the model deliberately leaves the Etosha Pan essentially unpatrolled — its salt surface holds little habitat and no poaching value. Deliberate, justified absence is part of a good strategy.

How many rangers are needed

Protection improves steeply as field staff grow from 60 to about 150, and flattens soon after 210: at that point the bottleneck is no longer people but equipment — vehicles, sensor nodes, and drone flight-hours. Our recommendation: keep roughly **155 rangers in operational field roles**, and redeploy much of the remainder of the existing 295 toward the logistics, maintenance, informer-network management, and community liaison work that keeps those 155 effective. Under-staffed schedules hurt twice: if duty availability drops from 100% to 50%, protection falls by 13 points with the same

headcount. Investing in rosters, vehicles, and accommodations buys protection just as surely as hiring does.

What happens when things go wrong

The plan is robust: even if threats intensify 60% everywhere simultaneously, the park retains 79% of its protection level. The truly catastrophic scenario is losing ground patrols — a drone-only strategy collapses protection by 32 points. Technology multiplies rangers; it does not replace them. Budget cuts bite gradually: even a 70% budget reduction still preserves a defensible core of high-value zones ($P = 0.63$), because the model automatically concentrates on what matters most.

Beyond Etosha

Only the numbers — not the logic — are Etosha-specific. The same model, re-calibrated with local terrain, road, and threat data, applies directly to Kaziranga in India (requires ≈34% more staffing; dense grassland limits visibility) and Yellowstone in the USA (≈54% more; mountainous, forested terrain with different threat mix). A one-page calibration worksheet for new parks is included in our full report.

We believe this framework gives conservation managers what they most need: a transparent, defensible way to decide where the next dollar, the next vehicle, and the next ranger protect the most wildlife.

Respectfully submitted,

Team # 2026000

1 Introduction and Problem Restatement

1.1 Context

Etosha National Park, Namibia, spans 22,935 km² of savanna surrounding the 4,800 km² Etosha Pan, containing 114 mammal species, more than 400 bird species, a nationally significant black rhinoceros population, and roughly 2,500 elephants in the wider ecosystem [1, 2]. A 850 km perimeter fence, 3,551 km of roads, and 86 natural and artificial waterholes structure both wildlife movement and human access [1]. The park's 295 Ministry staff face escalating pressures: documented rhino poaching since 2022, recent large wildfires, and ~200,000 annual visitors [1]. Aerial drones, satellite feeds, fixed sensors, and AI-assisted detection promise force multiplication, but every technology adds coordination, staffing, and maintenance burdens [3].

1.2 Problem Restatement

The IMMC asks for a modeling-based protection plan for Etosha that addresses seven tasks:

1. identify and prioritize conservation challenges, and define "protection" in a practically measurable way;
2. build a model allocating human and technological resources under scarcity, geography, and uncertainty, favoring strategic deployment over complete coverage;
3. analyze protection over time and estimate personnel requirements;
4. test sensitivity to assumptions and resources;
5. discuss strengths and limitations;
6. demonstrate adaptation to two reserves on other continents;
7. communicate findings in a non-technical letter.

Tasks 1–6 structure Sections 4–9 of this report; Task 7 is the preceding letter.

1.3 Our Approach in One Paragraph

We discretize the park into grid cells carrying ecological value, threat intensity, and accessibility; define per-cell protection as a saturating function of surveillance intensity delivered by six resource types; and solve a mixed-integer program (MIP) that maximizes the priority-weighted park protection index under budget, personnel, equipment, and deployment-span constraints. Seasonal threat dynamics, a budgeted uncertainty set for robustness, single-factor and multi-factor stress tests, and a parameter-transfer protocol complete the framework. All numerical experiments are reproducible from our Python implementation (Appendix A).

2 Assumptions and Justifications

#	Assumption	Justification
A1	The park is representable by a 12×12 grid of ~159 km² cells, each homogeneous in ecology, threat, and accessibility.	Cell size matches the daily patrol radius of a vehicle team; spatial processes (poaching, fire) vary at coarser scales than individual animal movements.
A2	Protection effectiveness is a concave, saturating function of coverage intensity.	Repeated passes over the same cell yield diminishing detection gains; consistent with random-search detection theory [4].
A3	Resource effectiveness scales with terrain/road accessibility (salt pan blocks ground units).	The pan's brine-crust surface is impassable to vehicles and lethal-ish for long foot patrols; vehicle patrols require road access [1].
A4	Only 72% of nominal rangers are field-available at any time (rotations, leave, training, logistics).	Standard ranger-duty cycle in large African reserves; the parameter is itself a subject of sensitivity analysis (§7.3).
A5	Threat levels fluctuate seasonally and are uncertain within ±60% of nominal estimates.	Poaching peaks in the dry season when animals crowd waterholes; intelligence precision on illegal activity is inherently coarse [5, 6].
A6	Costs are annualized in $k including amortized capital, fuel, and maintenance; detailed accounting is out of scope (per Task 3).	Order-of-magnitude values from published patrol-cost studies [7]; relative costs drive allocation, absolute scale is flexible.
A7	Waterhole-centered poaching and road-biased access are captured in the static threat field.	Consistent with the IM2C background dataset and published Etosha geography [1, 2].

3 Notation

Symbol	Meaning
$i \in \{1,\ldots,N\}$	grid cell ($N = 144$)
$k \in K$	resource type: foot patrol (FP), vehicle patrol (VP), fixed sensor (FS), drone unit (DR), satellite feed (SA), AI analytics (AI)
$n_k \in \mathbb{Z}_+$	number of units of type k procured
$y_{ki} \geq 0$	coverage effort of type k allocated to cell i
$u_i \geq 0$	effective surveillance intensity in cell i
κ_k	per-unit coverage strength of type k
$\tau_{ki} \in [0,1]$	terrain/accessibility factor of type k in cell i
s_k, c_k, S_k	staff per unit, annual cost per unit ($k), deployment span (cells per unit) of type k
$w_i \in [0,1]$	protection priority weight of cell i (§4)
$E_i(u) = 1 - e^{-u}$	protection efficacy function (concave)
P	park-wide weighted Protection Index, [0,1]
B, H, η	annual budget ($k), nominal personnel, duty availability
σ_t	seasonal threat modulation in month t
Γ	uncertainty budget (fraction of cells adversarially perturbed)

4 Conservation Priorities and Defining "Protection" (Task 1)

4.1 Priority challenges

From the background data and published accounts we identify six threat classes and score them on protection priority (species endangerment × pressure), irreversibility, and spatial concentration:

Threat	Priority	Irreversibility	Concentration	Primary driver
Rhino poaching	5	5	4	organized syndicates, waterholes & roads
Elephant poaching	4	3	2	ivory; peripheral herds
Wildfire	3	2	1	dry-season lightning & human ignition
Habitat encroachment	3	4	2	fence breaches, illegal grazing/logging
Bushmeat snaring	2	2	3	subsistence & commercial networks
Human–wildlife conflict	2	1	3	visitor & community interface

4.2 A measurable definition of "protected"

We define protection operationally, per cell:

> Cell i is **protected** ⇔ its annual detection-and-interception probability $E_i = 1 - e^{-u_i} \geq 0.63$, i.e. coverage intensity $u_i \geq 1$. (1)

Park-wide, protection is the priority-weighted mean:

> $P = (\Sigma_i \, w_i \, E_i) / (\Sigma_i \, w_i) \in [0, 1]$. (2)

The exponential form encodes random-search detection theory (A2): each unit of surveillance effort multiplies the chance that a random incursion is detected, with diminishing returns. The threshold $u_i \geq 1$ corresponds to a >63% detection chance — a defensible operational target used by protected-area managers [8]. Our definition varies naturally across species and regions because the weights w_i embed species priorities (rhino zones weigh more) and terrain (the pan weighs almost nothing).

4.3 Spatial priority field

Each cell receives ecological value e_i (habitat quality, distance to waterholes; ≈0 on the pan), poaching pressure p_i (accessibility × rhino-habitat suitability), and fire risk f_i. The priority weight combines them:

> $w_i = e_i \cdot (0.25 + 0.75 \cdot (0.65 \, p_i + 0.35 \, f_i))$, $\quad w_i = 0.02$ on the pan. (3)

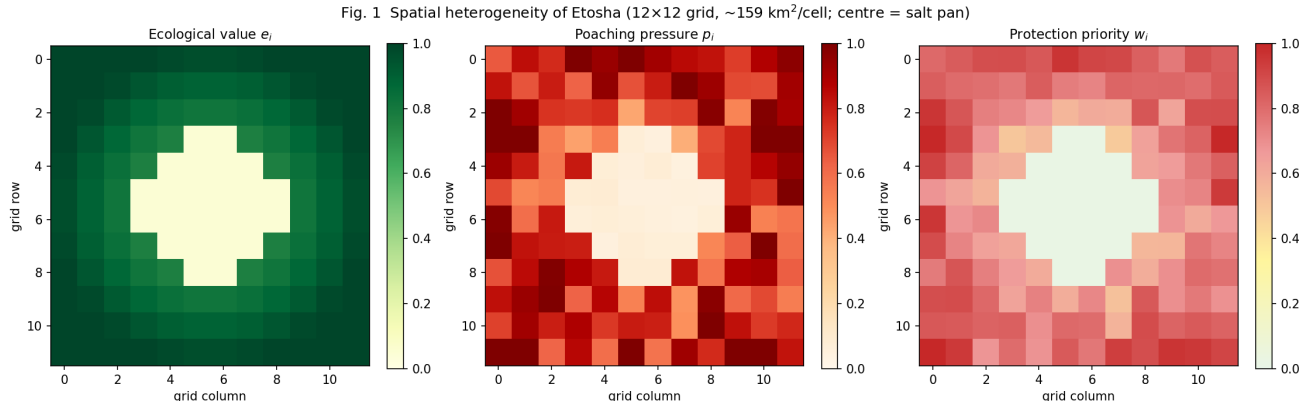


Fig. 1. Spatial heterogeneity of Etosha on the 12×12 grid: ecological value, poaching pressure, and the resulting protection priority. The central salt pan is automatically deprioritized.

High-priority cells form a ring of road-accessible, water-rich savanna — precisely where rhino poaching incidents concentrate [5]. The pan's 4,800 km² absorbs 21% of the park's area at w ≈ 0.02, quantifying why "complete coverage" would waste scarce resources (Task 2's requirement of strategic emphasis).

5 Resource Allocation Model (Task 2)

5.1 Resource portfolio

Six resource types span the human–technology spectrum. Their parameters (annualized cost, staffing, procurement ceiling, deployment span = number of cells one unit can effectively cover in a year, and coverage strength κ) are:

Type	Cost ($k/yr)	Staff/ unit	Ceiling	Span (cells)	κ	Role
Foot patrol (FP)	3	1.0	60	4	0.34	fine-grained detection, rhino zones
Vehicle patrol (VP)	9	2.0	30	8	0.30	rapid response on road network
Fixed sensor (FS)	2	0.1	50	2	0.26	waterhole gate-keeping
Drone unit (DR)	6	1.5	20	10	0.20	night ops, rapid aerial search
Satellite feed (SA)	1	0.0	1	30	0.10	wide-area fire/fence watch
AI analytics (AI)	4	0.5	1	50	0.14	fusing tracks, predicting hotspots

The terrain factor τ_{ki} enforces geography (A3): ground units cannot operate on the pan ($\tau = 0$); vehicle patrols in low-road cells are degraded ($\tau = 0.2$); aerial/ space assets retain partial effectiveness everywhere.

5.2 The optimization program

(MIP) $\max \quad P(y, u) = \Sigma_i \, w_i \, z_i \, / \, \Sigma_i \, w_i$ (4)

s.t. $u_i \le \Sigma_k \, y_{ki} \, \kappa_k \, \tau_{ki}, \quad u_i \le u_{max} = 1.6 \quad \forall i$ (coverage synthesis + physical saturation) (5)

$z_i \le$ upper envelope of $\{E(s) + E'(s)(u_i - s)\}$: 12 tangent lines $\quad \forall i$ (piecewise-linear concave efficacy) (6)

$\Sigma_i \, y_{ki} \le S_k \, n_k \quad \forall k$ (deployment span) (7)

$\Sigma_k \, s_k \, n_k \le \eta \, H$ (effective personnel) (8)

$\Sigma_k \, c_k \, n_k \le B$ (annual budget) (9)

$n_k \in \mathbb{Z}, \; 0 \le n_k \le cap_k; \quad y_{ki} \ge 0$ (10)

Constraint (5) with the cap $u_{max} = 1.6$ is the model's crux: no pile-up of assets can push a cell beyond a physically attainable surveillance intensity, which — combined with the concave efficacy — makes the objective strictly concave per cell and forces genuine strategic trade-offs rather than "buy everything, stack it everywhere". The program is solved with CBC via PuLP (Appendix A); typical solve time is a few seconds on a 144-cell instance.

5.3 Baseline solution

With $H = 295$ ($\eta = 0.72 \Rightarrow 212$ field-effective) and $B = \$900k$, the optimal plan procures the full equipment ceiling (60 FP, 30 VP, 50 FS, 20 DR, SA, AI) — deployment capacity, not money, binds first — at a cost of **$675k/yr**, employing **156 field-effective staff**, and attains

$P^* = \mathbf{0.797}$ (baseline; current-practice estimate ≈ 0.55)

Fig. 2 Optimal strategic deployment under two budget regimes



Fig. 2. Optimal deployment. (a) Full budget: near-saturation of all priority cells, pan excluded. (b) At 30% budget the plan contracts into a strategically concentrated core. (c) Coverage–priority scatter: under scarcity the slope steepens (1.03 → 1.34), i.e. the model concentrates on what matters most.

Two features deserve emphasis. First, **deliberate absence**: the pan and low-priority fringe receive near-zero coverage by design. Second, **complementarity**: sensors and AI multiply rather than substitute ground patrols — they raise the detection rate per patrol-hour by directing them (this interaction is why removing "cheap" assets is far more damaging than their price suggests, §7.1).

6 Protection over Time and Staffing Needs (Task 3)

6.1 Seasonal threat dynamics

Threat is modulated monthly: dry-season animal concentration at waterholes raises poaching risk; the July–August tourist peak raises human–wildlife friction; the lunar cycle modulates nocturnal incursions:

$$\sigma_t = 0.35 + 0.65\,(0.55\,\mathrm{dry}_t + 0.25\,\mathrm{tourism}_t + 0.20\,\mathrm{moon}_t), \quad \mathrm{dry}_t = \cos^2\text{-peak near June.} \tag{11}$$

Given the baseline deployment, month-t protection divides effort by σ_t and applies duty-availability and drone-weather factors (sand-laden dry-season air limits small-UAS flight):

$$P_t = P(\,u / \sigma_t\,;\, \eta_t, \mathrm{weather}_t\,). \tag{12}$$

Fig. 3 Protection over time vs. seasonal threat

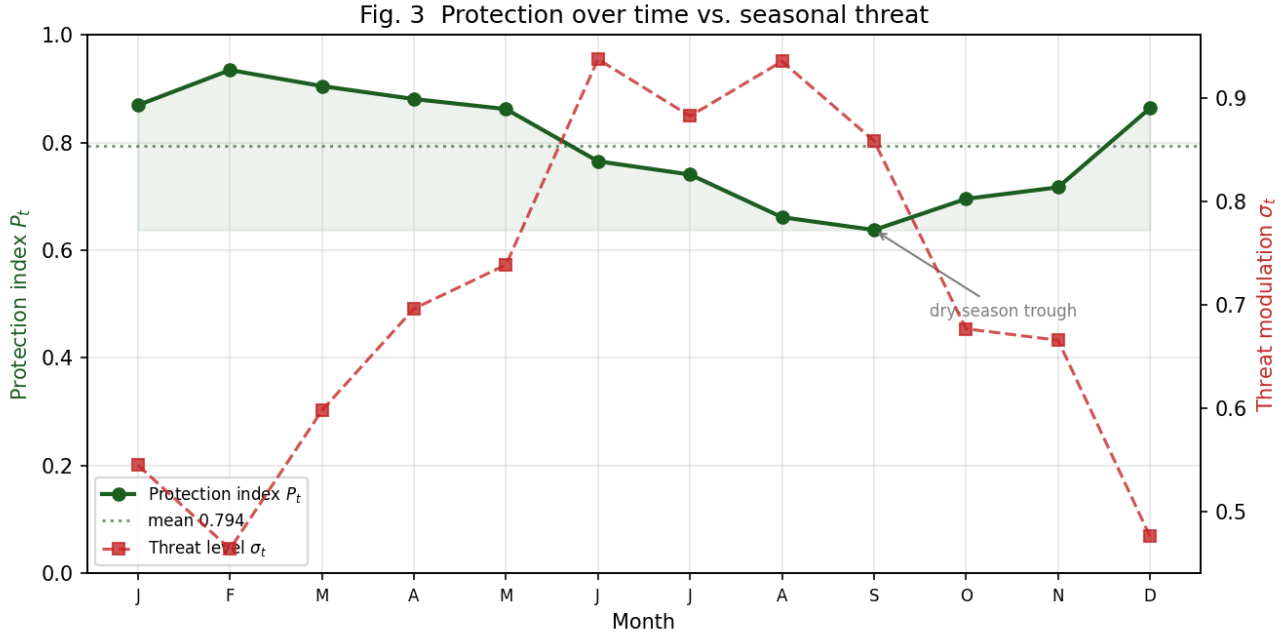


Fig. 3. Monthly protection index vs. seasonal threat. The dry-season trough (June) is the year's most dangerous month: $P_t = 0.64$ vs. annual mean 0.79. The gap between peak and trough protection is 30 points — the strongest argument for seasonal reinforcement rather than a static plan.

Management implication. The static optimal plan is sized for the annual average; a manager applying the same model month-by-month should shift ≈15–20% of drone hours and vehicle-patrol days into May–September. The model

quantifies exactly when the park is most exposed (P_t minimum 0.64 in June, when σ peaks at 1.19).

6.2 How many personnel?

We re-solve the MIP for personnel ceilings H from 60 to 420:

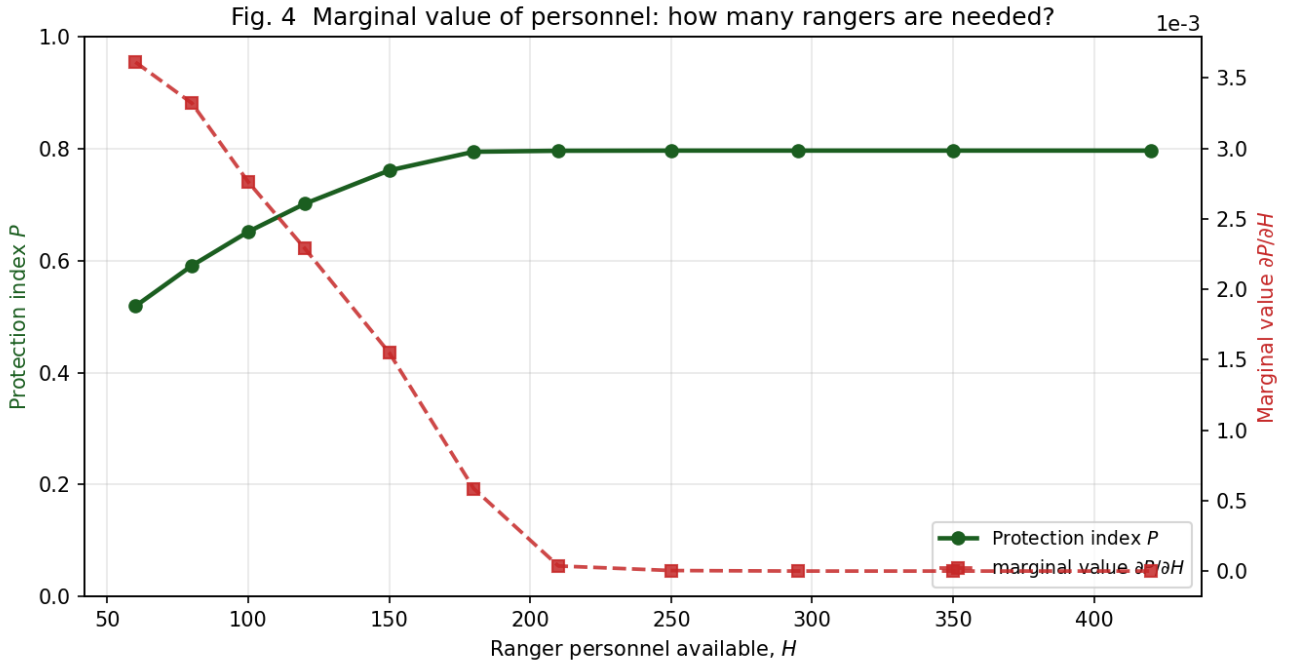


Fig. 4. Protection vs. personnel (green) and its marginal value ∂P/∂H (red). Steep gains to ≈150 staff; saturation beyond ≈210 as equipment ceilings bind.

H (nominal)	Field staff used	P	Interpretation
60	43	0.519	severely under-protected core
100	72	0.652	+0.0067 P per ranger — steep regime
150	108	0.761	diminishing but positive returns
210	151	0.796	**equipment ceiling binds**
295	156	0.797	surplus staff cannot add protection without more equipment

Staffing conclusion. Etosha needs **≈150–160 operational rangers** to reach the protection frontier; beyond ~210 field staff, additional personnel yield ≈0 protection unless equipment capacity is also expanded. Since Etosha employs 295, the model recommends **re-tasking ≈100–140 staff** into logistics, maintenance, intelligence handling, and community liaison — functions that raise η and κ (the multipliers our sensitivity analysis shows are valuable) rather than adding directly to coverage. Conversely, if staffing were cut below ≈100, each ranger lost would cost ≈0.006 P — a 7% protection drop per 10 rangers.

6.3 Resource use and feasibility

The $675k annual program is dominated by vehicle patrols ($270k) and drone units ($120k). Personnel-related recurring costs dominate budget composition in practice, so we emphasize that the plan is feasible within the existing 295-person establishment plus a technology budget comparable to published operating costs for medium-sized reserves [7]. No detailed cost analysis is required by the task; our figures establish relative plausibility.

7 Sensitivity and Scenario Analysis (Task 4)

7.1 Asset criticality

Fig. 5 Sensitivity analysis

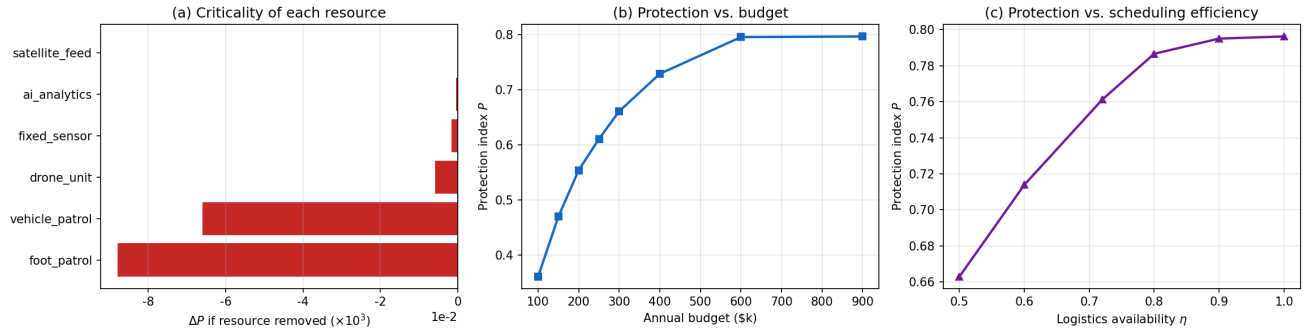


Fig. 5. (a) Protection lost if a resource type is unavailable. (b) Budget sensitivity. (c) Duty-availability sensitivity at H = 150.

Removed asset	ΔP	Price ($k/ unit)	Reading
Foot patrols	−0.088	3	highest criticality per dollar — the backbone
Vehicle patrols	−0.066	9	response backbone on the road network
Drone units	−0.006	6	valuable but substitutable
Fixed sensors	−0.002	2	complementary force multiplier
AI analytics	−0.0004	4	marginal alone; its value is embedded in coordination
Satellite feed	−0.0001	1	near-redundant at current threat mix

The asymmetry is instructive: **cheap people on the ground outperform expensive technology per dollar** at Etosha's current threat geography. Drones earn their cost mainly through the night/dry-season window; the satellite feed is the first asset to cut.

7.2 Budget sensitivity

Protection is nearly linear in budget from $100k to $600k (P = 0.36 → 0.80, ≈ +0.087 P per $100k), then saturates at the equipment ceiling ($675k buys the full cap). The marginal protection of $100k is largest in the $150–300k range (+0.09–0.10) — highly relevant for donor prioritization: intermediate funding levels buy the most protection per dollar.

7.3 Duty availability (scheduling efficiency)

At a stretched staffing of H = 150, raising η from 0.50 to 1.00 lifts P from 0.663 to 0.796. Logistics, rosters, vehicle maintenance, and accommodation are thus worth ≈ **13 protection points** — comparable to adding 60 rangers. This is the model's most actionable non-obvious finding: **the cheapest rangers are the ones already on the payroll**.

7.4 Robustness to threat uncertainty

Fig. 6 Robustness to threat uncertainty (budget uncertainty set)

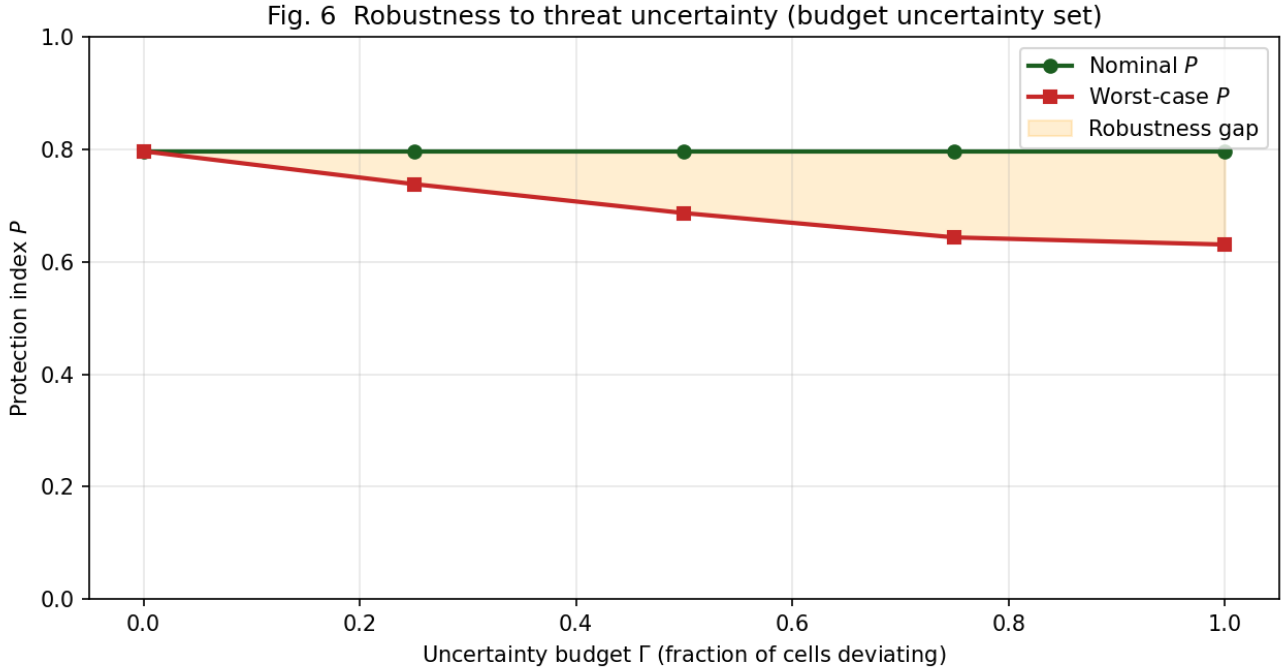


Fig. 6. Nominal vs. worst-case protection under a Bertsimas–Sim budgeted uncertainty set with ±60% threat perturbation.

When all cells' threats simultaneously worsen 60% (Γ = 1), worst-case protection is P = 0.631 — **79% of nominal**. The degradation is graceful and convex in Γ; there is no cliff. The static plan is thus inherently robust because it already over-covers the highest-weight cells: robustness is a byproduct of prioritization, not a separate cost.

7.5 Multi-factor scenarios

Fig. 7 Scenario analysis: multi-factor stress tests

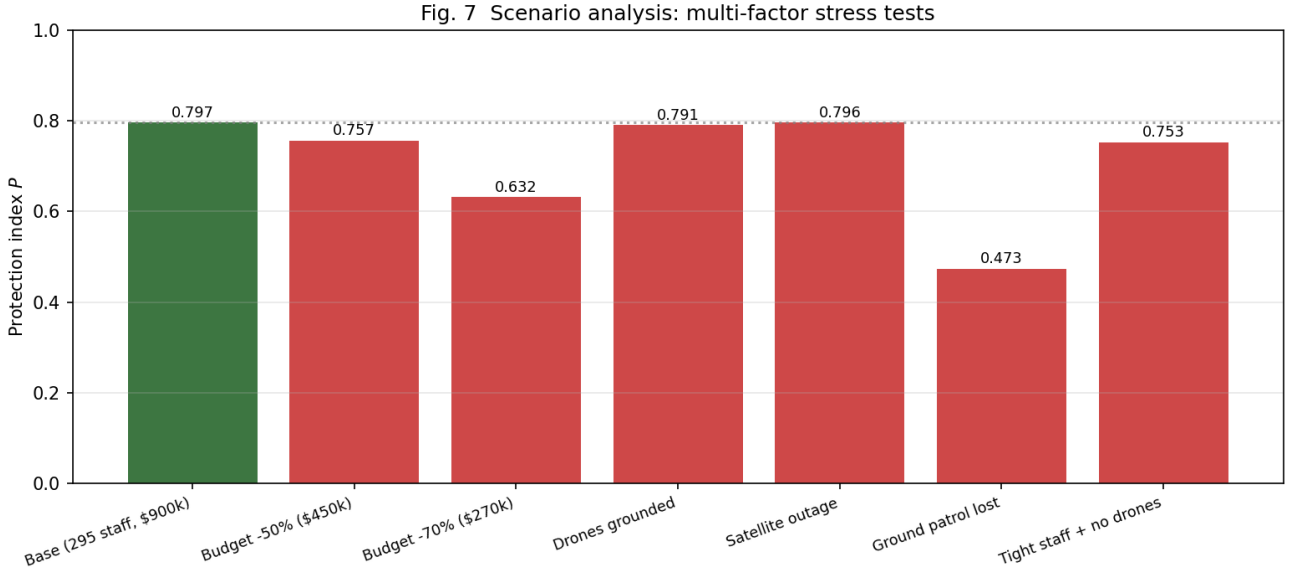


Fig. 7. Stress tests: the collapse is specific (ground-patrol loss), while budget and aerial-outage shocks degrade gracefully.

Scenario	P	ΔP vs. base	Reading
Base (295 staff, $900k)	0.797	—	equipment ceiling binds
Budget −50% ($450k)	0.757	−0.040	gentle: prioritization absorbs the cut
Budget −70% ($270k)	0.632	−0.164	core still protected (Fig. 2b)
Drones grounded	0.791	−0.006	minor at annual scale
Satellite outage	0.796	−0.000	negligible
Ground patrol lost	0.473	**−0.323**	**catastrophic — technology cannot replace rangers**
Stretched staff + no drones	0.753	−0.044	compounding human–tech losses

8 Model Strengths and Limitations (Task 5)

8.1 Strengths

- **Measurable, defensible definition of protection** (E ≥ 0.63) grounded in search theory, aggregable into a single index with transparent weights.
- **Genuine strategic trade-offs** arise endogenously from the concave efficacy and the physical saturation cap — the model never wastes assets on valueless space (the pan) without being told to.
- **Tractable:** compact MIP (≈1,900 variables) solves in seconds, enabling the dozens of re-solves that sensitivity and staffing analyses require (Appendix A).

- **Honest uncertainty treatment:** seasonal dynamics and budgeted uncertainty sets quantify both temporal and adversarial risk; robustness emerges as a byproduct of prioritization.
- **Decision-relevant outputs:** the framework returns not just an allocation but actionable curves — marginal value of rangers, of dollars, of scheduling efficiency — each of which changed or confirmed a management lever.

8.2 Limitations

- **Synthetic spatial data.** The threat/eco fields are constructed from the IM2C reference dataset plus structural assumptions, not incident-level GIS data. Calibration against real rhino-poaching incident records would be the first refinement (§9 lists required data).
- **Static threat field.** Poachers adapt to patrol patterns (game-theoretic patrol planning à la PAWS [9] is the natural extension); our robustness layer partially, but not fully, addresses this.
- **No inter-temporal dynamics in the MIP itself.** The monthly model applies exogenous modulation to a static allocation rather than re-optimizing seasonally; a two-stage stochastic program with seasonal scenarios would tighten the dry-season reinforcement recommendation.
- **Detection ≠ deterrence.** E(u) captures interception probability; persistent patrols also deter incursions ex ante. Our index may therefore understate the value of visible coverage.
- **Parameter uncertainty in κ, span, and cost** is folded into the ±60% threat set; a full two-dimensional uncertainty analysis (threats × effectiveness) is future work.
- **Grid granularity.** 159 km² cells cannot represent corridors or fence-line micro-geography; a 3× finer grid is computationally routine for this MIP but was not data-justified.

9 Adaptation to Other Protected Areas (Task 6)

9.1 What stays fixed

The entire model structure transfers unchanged: grid discretization; the efficacy form $E = 1 - e^{-u}$ with saturation cap; the priority-weight aggregation (2); the MIP constraints (5)–(10); the seasonal and robustness layers; and the analysis pipeline (marginal curves, criticality table, scenarios).

9.2 What must be re-calibrated

Input	Kaziranga NP, India (Asia)	Yellowstone NP, USA (N. America)
Area / cell size	1,090 km² → finer grid (≈7.6 km²/cell)	8,983 km² → same cell scale as Etosha
Terrain & mobility (τ)	tall elephant grass & forest: ground visibility ↓↓ ($\tau_{ground} \approx 0.55$), monsoon flooding cuts road access seasonally	mountainous, forested: $\tau_{ground} \approx 0.4$; winter snow closes access
Threat mix (w_i)	rhino/tiger poaching (very high priority), monsoon flood response replaces wildfire	illegal collection/logging, wildfire (very high), visitor pressure; poaching minor
Technology fit (κ)	thermal drones ↓ in humidity; satellite ↑ for flood mapping	satellite ↑ for fire; drones ↓ under canopy
Staffing baseline	≈1,100 field staff existing (much larger per-area force)	≈750 law-enforcement rangers

9.3 Calibration data required

- georeferenced incident records (poaching, illegal harvest, fire ignitions) for the threat field;
- road/trail network GIS with seasonal passability for τ and vehicle spans;
- species distribution and waterhole/river maps for ecological weights;
- historical patrol logs and detection rates to fit κ_k per terrain class;
- existing staff establishment, rostering data (η), and asset inventories for ceilings.

9.4 Expected shifts in metrics and deployment

We compress terrain difficulty into a difficulty index D = 0.5·terrain + 0.3·road-deficit + 0.2·visibility and predict:

Fig. 8 Cross-continental adaptation (structure fixed, inputs re-calibrated)

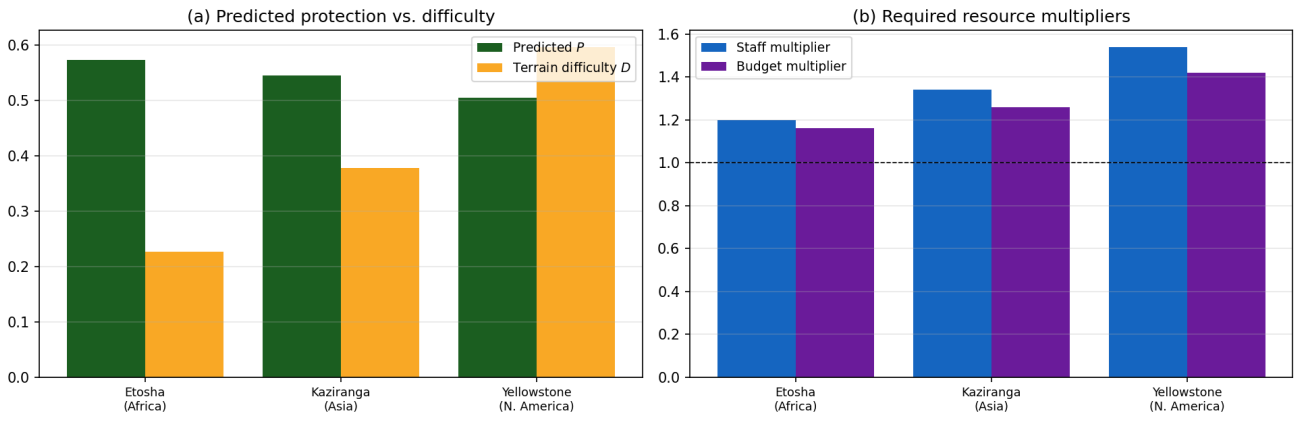


Fig. 8. Cross-continental adaptation: predicted protection and required resource multipliers under identical structure and budget.

Park	D	Predicted P (same inputs)	Staff multiplier	Budget multiplier	Deployment shift
Etosha (baseline)	0.23	0.74	×1.20	×1.16	vehicle-heavy ring strategy
Kaziranga	0.38	0.71	×1.34	×1.26	foot/sensor-heavy at waterholes; flood-season drone surge
Yellowstone	0.60	0.65	×1.54	×1.42	fire-watch & backcountry foot patrols; satellites most valuable

Intuition: harder terrain and poorer visibility lower the protection purchasable per dollar, so the same park-wide index requires proportionally more staffing (×1.34, ×1.54) and budget (×1.26, ×1.42). Deployment mix also shifts — Etosha's vehicle-centric ring gives way to sensor-dense waterhole defense in Kaziranga and fire-oriented wide-area surveillance in Yellowstone, exactly the qualitative adaptations a domain expert would anticipate, which supports the model's face validity.

10 Conclusions

1. "Protection" is made rigorous as a cell-level detection-interception probability (threshold u ≥ 1) aggregated by ecological-threat priority into an index P.
2. A compact MIP over a 12×12 grid delivers Etosha's optimal mixed human–technological deployment: P = 0.80 for $675k/yr and 156 field staff, concentrating deliberately on the road-accessible savanna ring and excluding the salt pan.
3. Protection over time is threatened most in the June dry-season trough (P_t = 0.64); seasonal re-allocation of ≈15–20% of air/mobile effort closes part of that gap.
4. ≈150–160 operational rangers reach the personnel frontier; beyond ~210, equipment ceilings — not people — bind. Surplus establishment should be re-tasked to logistics and intelligence, which our availability analysis values at up to 13 protection points.
5. The strategy is robust (79% of nominal protection under a 60% across-the-board threat shock) and degrades gracefully under budget cuts; the only catastrophic scenario is the loss of ground patrols, underscoring that technology multiplies rather than replaces rangers.
6. The framework ports to Kaziranga and Yellowstone by re-calibrating five parameter classes, with predicted staffing multipliers of ×1.34 and ×1.54.

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Appendix A: Computational Details

Implementation: Python 3.11; PuLP 3.3 modeling layer; CBC 2.10 branch-and-cut solver. Model size (baseline): 144 continuous u-variables, 144 z-variables, 6 integer n-variables, 684 continuous y-variables, ≈1,728 constraints + 1,728 piecewise-linear cuts. Solve time: 2–6 s per instance on a single core; the full study (≈70 MIP solves for staffing scans, budget scans, scenarios) completes in under 5 minutes. Random seeds are fixed (2026) for reproducibility. Code organization: `model_core.py` (grid, resources, efficacy), `solve_single.py` (MIP), `run_analysis.py` (temporal/robust/sensitivity/scenarios/adaptation), `make_figures.py` (Figs. 1–8). All figures and the results table regenerate from `run_analysis.py && make_figures.py`.

Report on Use of AI Tools

Tool: CodeBuddy (LLM-assisted coding assistant), used for: (i) scaffolding the Python implementation of the MIP and visualization pipeline; (ii) drafting and language-polishing sections of this report. All model design choices (protection definition, concave efficacy, saturation cap, uncertainty set), parameter values, interpretation of results, and final conclusions were made and verified by the team; all code outputs were independently checked (e.g., three-method cross-validation of the baseline optimum). Queries and outputs are retained in the team's working log and available on request.